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Roll no:.....

Degree : B.Tech.Semester:III-Course Work

END SEMESTER EXAMINATION (SUMMER),July,2026

Course Code :ECMTC04

Course Title:Mathematics for Signal Processing and Communication

Time:03 hours

Marks:50

**Note: Attempt all five questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.**

| Q. No. | Questions                                                                                                                                                                                                                                                                                                                                     | Marks | CO  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| Q1     | Attempt any 2 parts of the following:                                                                                                                                                                                                                                                                                                         |       |     |
| 1(a)   | Let $G = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{Z} \right\}$ be a group under matrix addition. Let $H = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in G : 2a - 3d = 0 \right\}$ Prove that H is a subgroup of G. What happens if the condition $2a - 3d = 0$ is replaced by $2a - 3d = 2$ ? | 5     | CO1 |
| 1(b)   | Let $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \cdot)$ denote the additive and multiplicative groups, respectively. Show that the map $\phi: \mathbb{R} \rightarrow \mathbb{R}^+$ given by $\phi(x) = e^x$ is a group homomorphism. Also, determine the kernel of $\phi$ .                                                                         | 5     | CO1 |
| 1(c)   | Define the term <b>field</b> . Determine whether the ring $\mathbb{Z}_{11}$ is a field. Justify your answer. Also comment on whether the ring $\mathbb{Z}_{12}$ is a field or not.                                                                                                                                                            | 5     | CO1 |
| Q2     | Attempt any 2 parts of the following:                                                                                                                                                                                                                                                                                                         |       |     |
| 2(a)   | Define the term subspace of a vector space. Let $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + 3z = 0\}$ . Determine whether W is a subspace of $\mathbb{R}^3$ . Justify your answer.                                                                                                                                                           | 5     | CO2 |
| 2(b)   | Define linearly independent and linearly dependent vectors. Determine whether the vectors $(1,0,2), (2,1,3), (3,1,5)$ form a basis for $\mathbb{R}^3$ over $\mathbb{R}$ .                                                                                                                                                                     | 5     | CO2 |
| 2(c)   | Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y - z)$ . Show that T is a linear transformation. Determine $\dim(\ker T)$ and hence check the invertibility of T                                                                                                                                          | 5     | CO2 |
| Q3     | Attempt any 2 parts of the following:                                                                                                                                                                                                                                                                                                         |       |     |

|      |                                                                                                                                                                                                                                                                   |   |     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|
| 3(a) | If the directional derivative of $\phi = axy + byz + czx$ at $(1,1,1)$ has maximum magnitude 4 in a direction parallel to $x$ axis, find the values of $a, b, c$ .                                                                                                | 5 | CO3 |
| 3(b) | For constant vector $\vec{a}$ , show that<br>(i) $\nabla (\vec{a} \cdot \vec{r}) = \vec{a}$<br>(ii) $\nabla \times (\vec{a} \times \vec{r}) = 2\vec{a}$                                                                                                           | 5 | CO3 |
| 3(c) | Verify Green's theorem for the field $\vec{F} = x^2\vec{i} + xy\vec{j}$ over the region $R$ enclosed by $y = x^2$ and the line $y = x$ .                                                                                                                          | 5 | CO3 |
| Q4   | Attempt any 2 parts of the following:                                                                                                                                                                                                                             |   |     |
| 4(a) | If $f(z) = u + iv$ is an analytic function of $z$ , then find $f(z)$ if $u - v = (x - y)(x^2 + 4xy + y^2)$ .                                                                                                                                                      | 5 | CO4 |
| 4(b) | Evaluate $\int \frac{\sin^6 z}{(z - \pi/2)^3} dz$ around the circle $ z  = 2$ .                                                                                                                                                                                   | 5 | CO4 |
| 4(c) | Find the image of the strip $-\frac{\pi}{2} < x < \frac{\pi}{2}, 1 < y < 2$ under the mapping $w(z) = \sin z$ .                                                                                                                                                   | 5 | CO4 |
| Q5   | Attempt any 2 parts of the following:                                                                                                                                                                                                                             |   |     |
| 5(a) | A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and the temperature initially is<br>$U(x,0) = x, \quad 0 \leq x \leq 50$ $= 100 - x, \quad 50 \leq x \leq 100$<br>Find the temperature $u(x,t)$ at any time $t$ . | 5 | CO5 |
| 5(b) | A string is stretched and fastened to two points $L$ apart. Motion is started by displacing the string in the form $u = a \sin \frac{\pi x}{L}$ from which it is released at time $t = 0$ . Find the displacement $u(x,t)$ from one end.                          | 5 | CO5 |
| 5(c) | Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the conditions<br>$u(0,y) = u(l,y) = u(x,0) = 0$ and $u(x,a) = \sin\left(\frac{n\pi x}{l}\right)$ .                                                             | 5 | CO5 |